

Fig. 7. Seaplane landing on the ocean. A seaplane with a 10 m wingspan, pitched 9° nose-up, approaches at 8 m/s, touches down, and skims the surface at 10 m/s, leaving a long trailing wake that transmits smoothly across the 2D–3D interface. In each panel, the main image is the far view and the inset is the near view of the same moment. The wave model uses $\varepsilon = 1$.

Algorithm 1 ComputeVolumeVelFrom2D (bulk velocity from (η, ψ) via the Craig–Sulem extension and finite-depth kernels)

Input: surface state (η, ψ) on wave grid, kernels $\{C_i, S_i\}_{i=0}^{L_d-1}$, padding mode Pad.
Output: layered velocities $\{(u_i, v_i, w_i)\}_{i=0}^{L_d-1}$.

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1: if Pad then
2:    $(\hat{\eta}, \hat{\psi}) \leftarrow \text{EvenExtend}(\eta, \psi)$  ▷ Sec. 3.3
3: end if
4:  $\hat{\eta}, \hat{\psi} \leftarrow \text{FFT}(\eta), \text{FFT}(\psi)$ 
5:  $\hat{\Phi}|_{y=0} \leftarrow \hat{\psi} - \varepsilon (\hat{\eta} \cdot G_0 \hat{\psi})_{3/2}^\wedge$  ▷ Eq. (21)
6:  $\hat{\Phi}|_{y=0} \leftarrow \hat{\Phi}|_{y=0} \cdot \text{taper}_{\text{couple}}$  ▷ Sec. 6.2
7: for  $i = 0, \dots, L_d - 1$  do
8:    $\hat{u}_i, \hat{w}_i \leftarrow ik_x C_i \hat{\Phi}|_{y=0}, ik_z C_i \hat{\Phi}|_{y=0}$ 
9:    $\hat{v}_i \leftarrow S_i \hat{\Phi}|_{y=0}$ 
10:   $(u_i, v_i, w_i) \leftarrow \text{IFFT}(\hat{u}_i, \hat{v}_i, \hat{w}_i)$ 
11:  if Pad then  $(u_i, v_i, w_i) \leftarrow \text{Crop}(u_i, v_i, w_i)$ 
12:  end if
13: end for
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model is canonical in (η, ψ) but the 3D NS solver tracks a level set ϕ and a velocity field $\mathbf{u}$; there is no canonical ψ directly available on the 3D side, so the wave model must reconstitute it. Second, naive overwriting of the wave state inside the box would create a discontinuity at the box edge that radiates spurious reflections back into the open ocean. We resolve the first issue by transferring only the surface elevation η from the 3D solver and reconstructing the canonical ψ on the wave side through an inverse DNO solve $\psi = G(\eta)^{-1}v$, which converts the pair (η, v) back into the canonical conjugate, so that no fluid-derived source acts on the dynamic equation for ψ . The second issue is handled by formulating the data transfer as an implicit relaxation source on the Zakharov right-hand side that decays exponentially through a smooth transition band.

5.1 Kinematic targets from the 3D state

Let $\phi^{3D}(\mathbf{x}, y, t)$ be the 3D level set at the current substep, with y the vertical axis and (x, z) the horizontal plane of the wave grid. We construct the elevation target $\eta^{\text{target}}(x, z)$ on the wave grid by a column-wise scan of ϕ^{3D} , following the strategy of Chentanez et al. [2015]. For each wave-grid column (i, k) , we sweep the 3D cells

from the bottom upward and count consecutive air/solid cells; once the running count reaches a small threshold n_{air} (we use $n_{\text{air}} = 3$), we identify the cell n_{air} steps below as the surface cell j^* , and set

$$\eta_{i,k}^{\text{target}} = (j^* + \frac{1}{2} + o_y) \Delta x - \phi_{i,j^*,k}^{3D}, \quad (30)$$

i.e., the y -coordinate of the surface cell center, corrected by the signed-distance value at that cell as a sub-cell offset (o_y is the box origin offset in y). To keep this target compatible with the wave grid, we then apply a few iterations of FAB-band padding plus a small Laplace smoothing to η^{target} ; this damps the high-frequency staircase noise that column-scanned level sets exhibit.

5.2 Relaxation source on the Zakharov system

With the elevation target η^{target} in hand, we drive the wave state toward the 3D state by augmenting the Zakharov system with a relaxation source on the canonical equations,

$$\partial_t \eta = G(\eta) \psi - \sigma(\mathbf{x})(\eta - \eta^{\text{target}}), \quad (31)$$

$$\partial_t \psi = -g\eta + \mathcal{N}_\psi, \quad (32)$$

in which $\sigma(\mathbf{x}) \geq 0$ is a spatially graded relaxation rate (units 1/s) supported on the embedding-box footprint. We choose

$$\sigma(\mathbf{x}) = \sigma_{\text{max}} \rho \left(\frac{\text{dist}(\mathbf{x}, \partial\Omega_{3D})}{w_b} \right) \mathbf{1}_{\mathbf{x} \in \Omega_{3D}}, \quad (33)$$

where $\partial\Omega_{3D}$ is the box edge, w_b is the transition-band width, $\rho : [0, 1] \rightarrow [0, 1]$ is a smooth ramp ($\rho(0) = 0, \rho(1) = 1$), and $\mathbf{1}_{\mathbf{x} \in \Omega_{3D}}$ is the indicator of the box footprint, which restricts the support of σ to the box interior. Outside the box, $\sigma(\mathbf{x}) \equiv 0$ and the wave evolves freely. In the transition band σ rises from 0 to σ_{max} , and in the box interior $\sigma \equiv \sigma_{\text{max}}$ pulls the wave state toward the 3D state at rate σ_{max} . Here, the rate σ is a property of position, not of the wave field, so the source enters (31) *linearly*; this is what makes it amenable to exact integration.

The source acts on the kinematic equation (31) alone because (η, ψ) are canonically conjugate rather than two independent fields: the wave Hamiltonian already commits ψ 's evolution to η through the dynamic equation (32), so pinning η to η^{target} already determines what ψ must do, and adding an independent ψ^{target} source would risk over-determining the pair and breaking the symplectic identity $v = G(\eta)\psi = \partial_t \eta$ inside the box. In practice such a fluid-derived target carries the NS solver's numerical viscosity and discretization

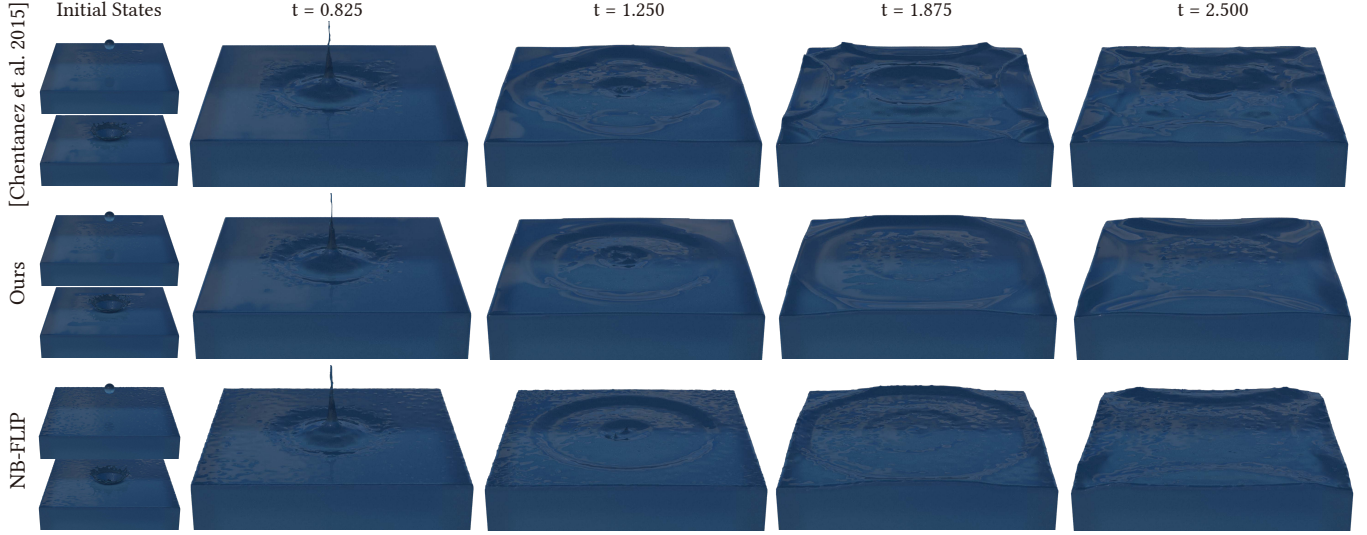


Fig. 8. Crown Splash. The crown splash scenario compares Chentanez et al. [2015] (top row), ours (middle row), and the 3D NB-FLIP reference (bottom row). Our method produces results that closely match the NB-FLIP reference, faithfully preserving crown splash dynamics and enabling smooth wave transmission across the 2D–3D interface. In contrast, the method of Chentanez et al. [2015] exhibits clearly incorrect wave propagation in the 2D region, as the shallow water equations cannot reproduce the dispersive behavior of the deeper-water reference. Our wave model uses $\varepsilon = 0.8$.



Fig. 9. A surfacing submarine advancing forward and slowly rising to the ocean surface. In each panel, the main image is the far view and the inset is the near view of the same moment. The wave model uses $\varepsilon = 0.3$.

artifacts, so it generally disagrees with the conjugate the wave Hamiltonian produces, and the mismatch radiates from the box edge. We validate this choice empirically in Figure 17: additionally driving ψ toward a Helmholtz-decomposed target extracted from the 3D surface velocities produces visibly degraded results.

5.3 Implicit integration of the relaxation source

A naive explicit treatment $\eta_{n+1} = \eta_n - \sigma \Delta t (\eta_n - \eta^{\text{target}})$ becomes unstable for $\sigma_{\max} \Delta t > 2$, which would force a uselessly small Δt when the box must absorb a violent feature. Because the source is linear in the deviation $\eta - \eta^{\text{target}}$, the relaxation has a closed-form solution: holding η^{target} frozen across one substep, the relaxation flow is

$$\eta(t + \Delta t) = \eta^{\text{target}} + (\eta(t) - \eta^{\text{target}}) e^{-\sigma(x) \Delta t}, \quad (34)$$

which is unconditionally stable, monotonically convergent, and exact at every σ . We compose this analytic decay with the wave step of Section 3.4 via operator splitting: first advance (η, ψ) by Δt under the wave Hamiltonian, then apply (34) pointwise.

Equation (34) updates η but says nothing about the wave’s surface vertical velocity $v = G(\eta)\psi$, and we deliberately do not introduce a

target v^{target} to drive it with. At every substep we preserve the wave’s own v across the η update (the value $v = G(\eta_{\text{step}})\psi_{\text{step}}$ carried out of the Hamiltonian substep) and recompute the canonical conjugate from the post-relaxation pair $(\eta^{\text{relaxed}}, v)$ via an inverse DNO solve,

$$\psi^{\text{relaxed}} = \left[G(\eta^{\text{relaxed}}) \right]^{-1} v, \quad (35)$$

see (19). The triple (η, ψ, v) is therefore self-consistent at the end of each substep.

We do not introduce a v^{target} because the information v carries is already encoded in the time history of η^{target} : the canonical relation $v = G(\eta)\psi$ together with the unmodified Hamilton evolution of ψ means that pinning η to η^{target} across substeps automatically pulls the wave’s v toward $\partial_t \eta^{\text{target}}$, so a separate v target would supply little independent information. The single-boat experiment in Figure 12 also supports this choice: variant (c), which additionally relaxes v toward a finite-difference target built from the 3D level set, introduces spurious reflections behind the stern, while variant (e), which relaxes η alone, lets the outgoing wake pass through cleanly.



Fig. 10. Two chasing boats on the ocean surface, producing long trailing wakes as they move through the water. The four images on the left show the near view, while the large image on the right shows the far view. The wave model uses $\varepsilon = 1$.

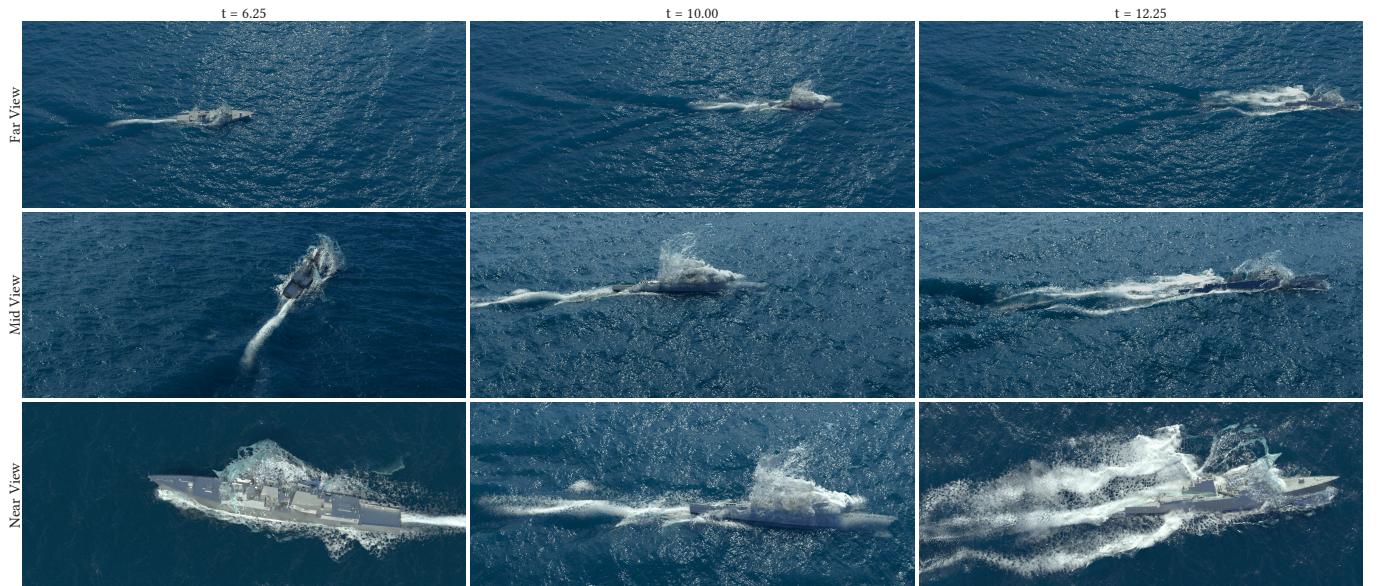


Fig. 11. Battleship sailing through heavy waves under weak two-way solid–fluid coupling. The 2D wave domain measures 120 m×60 m and embeds a 24 m×12 m×12 m 3D fluid box around the hull. Columns correspond to three time instants ($t = 6.25, 10.00, 12.25$ s); the rows show, from top to bottom, the far, mid, and near views of the same scene. The simulation remains stable under the near-breaking background spectrum, and the 2D and 3D regions stay consistent in the frames shown; residual boundary artifacts visible in the accompanying animation are discussed in Section 8. The wave model uses $\varepsilon = 0.2$.

6 Implementation Details

Our method is implemented in Warp [Macklin 2022] with GPU acceleration. In 2D simulation, we utilize PyTorch’s FFT module and padding functions to handle spectral operations and boundary conditions efficiently. Our 2D solver is compatible with standard ocean spectrum initializations. In some of our experiments, the initial height and velocity are generated from a TMA spectrum with Donelan–Banner directional spreading, following Horvath [2015]. For 3D simulation, we implement NB-FLIP [Ferstl et al. 2016] with weak two-way solid–fluid coupling following [Bridson 2015]. Solid objects are represented in voxelized cells. The simulation runs on staggered grids. Pressure projection is solved with a CUDA-based

unsmoothed aggregated Algebraic Multigrid Preconditioned Conjugate Gradient solver [Shao et al. 2022; Sun et al. 2025]. Whitewater (spray, foam, and bubbles) shown in our results is simulated as a post-process using Houdini’s built-in whitewater solver, driven by the velocity and surface fields produced by our simulation. The fine-scale ripples and shimmering visible on the ocean surface in our rendered results are small ocean-spectrum displacements added in Houdini at render time for visual richness; they are disabled in all quantitative comparisons.

Masking residual coupling artifacts. While our coupling suppresses most interface artifacts, it does not eliminate them entirely. In our most demanding scene, the battleship in heavy seas (Figure 11),

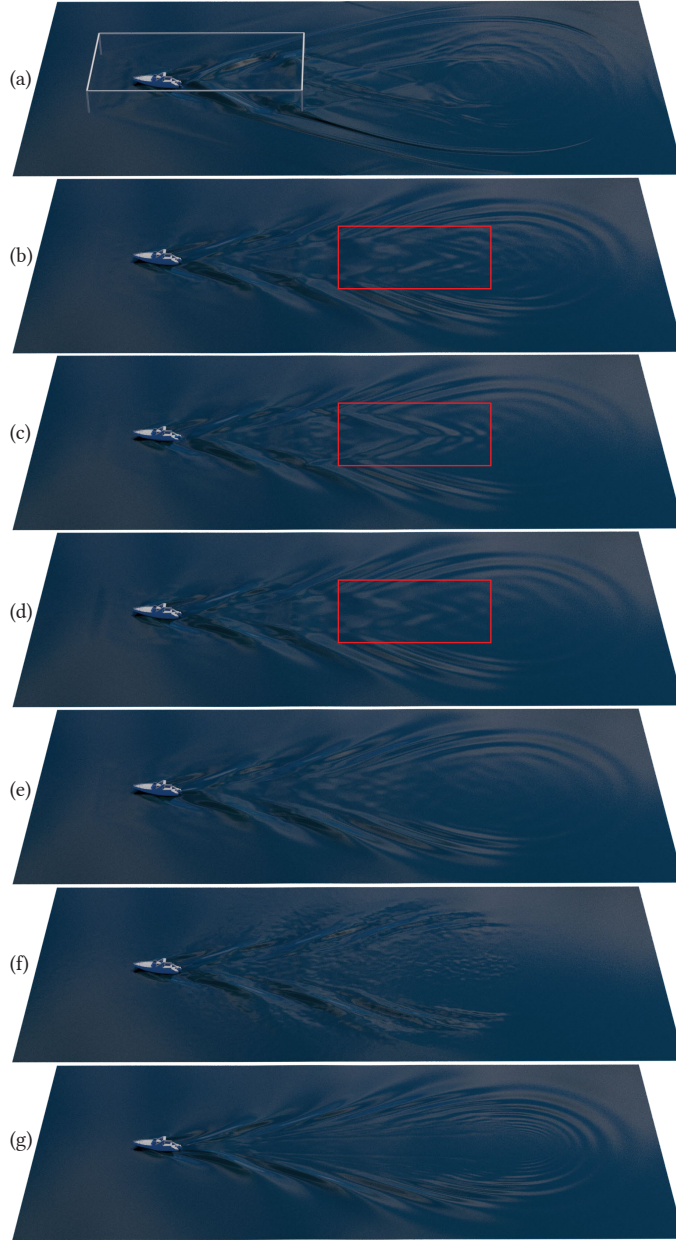


Fig. 12. Single-boat wake comparison across seven methods. (a) Chentanez et al. [2015] (the white box marks the common 3D simulation region): the wake is qualitatively wrong, as the shallow water equations cannot resolve the dispersive Kelvin pattern. (b) Airy DK [Canabal et al. 2016] coupled via the FAB scheme [Stomakhin and Selle 2017], (c) coupling through height and velocity relaxation, and (d) our wave model combined with the coupling scheme of Chentanez et al. [2015] all reproduce the feather-like wake [Rabaud and Moisy 2013] but exhibit clear spurious reflections behind the stern (highlighted by the red boxes). (e) Our full method reproduces the feather-like Kelvin wake with minimal spurious reflection at the 2D–3D interface. (f) 3D NB-FLIP reference: the trailing wave at the far right is noticeably damped due to numerical dissipation of the 3D solver. (g) Pure 3D Eulerian level-set simulation. The variants built on our wave model, (c), (d), and (e), use $\varepsilon = 1$.

the accompanying animation shows wavefronts reflecting from the rectangular boundary of the 3D box and propagating into the 2D domain. In production settings, such residual artifacts can be effectively obscured using two standard render-time layers. First, the ocean-spectrum displacements described above can be amplified to overlay additional procedural spectral detail on the simulated surface. Second, foam and whitewater can be emitted to further obscure the remaining artifacts.

6.1 De-aliased nonlinear evaluation

Because every G_j for $j \geq 1$ is built from quadratic and higher-order pointwise products in η , $\nabla\eta$, $\nabla\psi$, $G_0\psi$, and $B_0\psi$, naive spectral evaluation aliases high-wavenumber content back into the resolved band and corrupts the nonlinear corrections within a few steps. We follow Orszag’s 3/2-rule [Orszag 1971]: every quadratic product in (7), (8), and (10) is computed by zero-padding both factors from (M, N) to $(\frac{3}{2}M, \frac{3}{2}N)$, multiplying in physical space on the padded grid, and truncating the result back to (M, N) . The padded grid is large enough that the highest aliased mode $|\mathbf{k}| + |\mathbf{k}'| = 2|\mathbf{k}|_{\max}$ falls outside the retained band of the truncated product, so the surviving modes are exact. After de-aliasing we apply a smooth super-Gaussian taper $\exp[-(|\mathbf{k}|/k_c)^8]$ with $k_c = \frac{2}{3}k_{\text{Nyq}}$ to all nonlinear products. The taper is unity through most of the resolved band and prevents the high-wavenumber tail of G_2 , whose symbol grows like $|\mathbf{k}|^2$, from amplifying numerical noise injected by the coupling boundary.

6.2 Spectral bandlimiting of the boundary condition

The kernels in (27)–(29) act as spectral derivatives, so they amplify any high-wavenumber content in $\hat{\Phi}|_{y=0}$ linearly in $|\mathbf{k}|$. That tail is also the least reliable part of the wave field, since the truncation of Craig–Sulem expansion loses accuracy near the wave grid’s Nyquist. Feeding it unaltered into the 3D solver injects near-grid-scale forcing that the pressure projection then has to fight. We therefore multiply $\hat{\Phi}|_{y=0}$ by a separable super-Gaussian taper $\exp[-(|k_x|/(\frac{1}{2}k_{\text{Nyq}}^\alpha))^8]$ for $\alpha \in \{x, z\}$ before applying (27)–(29). The taper is unity through the lower half of the spectrum and decays smoothly to suppress the upper half, thereby preserving the resolved nonlinear wave content while keeping the 3D pressure solve free of injected high-wavenumber noise.

7 Time Integration

At each substep the 3D solver advances by Δt_{3D} , set by an outer CFL on the 3D advection and solid motion. The 2D wave solver uses its own step $\Delta t_{2D} \leq \Delta t_{3D}$: in most of our scenes we simply take $\Delta t_{2D} = \Delta t_{3D}$, but when finer resolution of the surface dynamics is desired we sub-cycle the wave solver with a smaller Δt_{2D} within one 3D step. The main algorithm is summarized below and outlined in Alg. 2. In the enumerated description that follows, the parenthetical labels refer to the corresponding line numbers of Alg. 2.

- (1) **Compute Time Step (Step 1).** Based on the CFL condition on the 3D velocity field and the solid motion, we compute the substep size Δt .
- (2) **Advect 3D (Step 2).** We perform the 3D level-set advection and velocity transport as in standard NB-FLIP, producing the tentative state $(\phi_{3D}^{n+1}, \mathbf{u}_{3D}^{n+1})$.

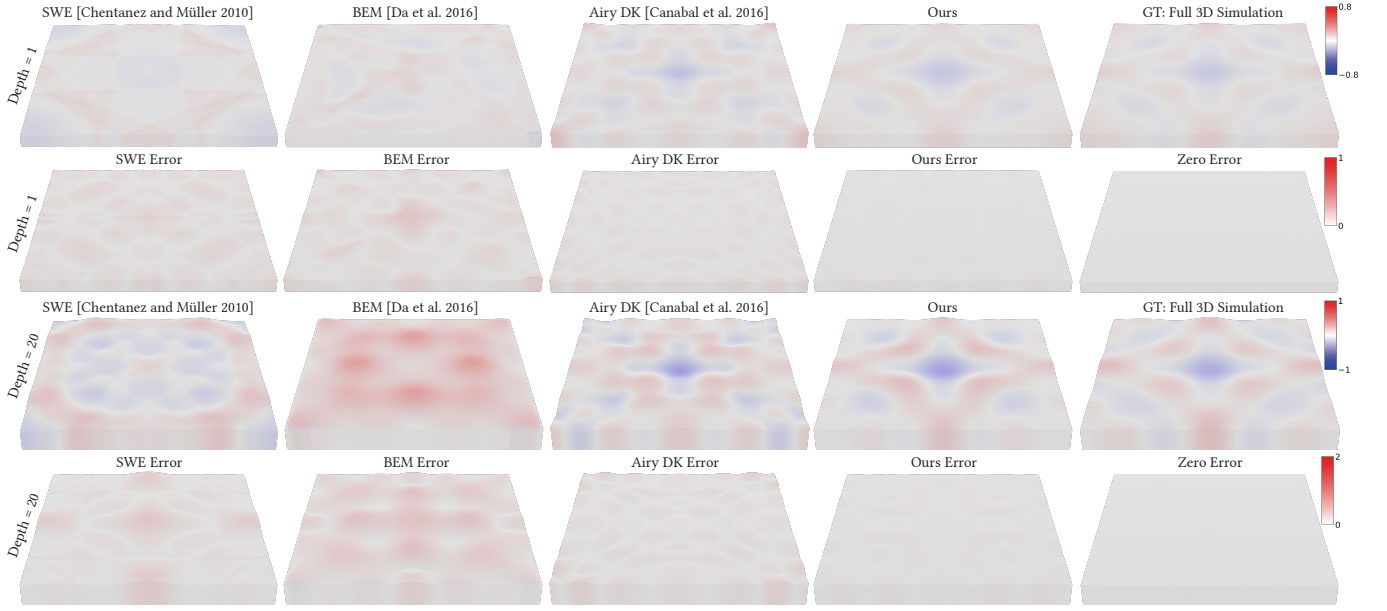


Fig. 13. Wave propagation comparison in a 20-meter-wide tank under shallow (depth = 1 m) and deep (depth = 20 m) water conditions. For each depth, the upper row shows the rendered free surface produced by SWE [Chentanez and Müller 2010], BEM [Da et al. 2016], Airy DK [Canabal et al. 2016], our nonlinear wave model ($\varepsilon = 1$), and a high-resolution 3D Eulerian level-set simulation that serves as the ground-truth reference; the lower row shows the corresponding per-cell absolute height error against this 3D reference. Our wave model agrees most closely with the 3D reference in both regimes.

- (3) **Update Solid (Step 3).** We advance the rigid body by Δt using its current linear and angular velocities and refresh the solid–fluid information carried by the 3D grid (solid SDF samples, solid velocity, and cell-type tags).
- (4) **Prepare Targets (Steps 4–5).** We build η^{target} from $\tilde{\phi}_{3D}^{n+1}$ by the column-wise scan: sweep each column upward until $n_{\text{air}} = 3$ consecutive air/solid cells appear, then read the surface from $\tilde{\phi}_{3D}^{n+1}$ at the cell n_{air} steps below (Eq. 30, Section 5.1), followed by FAB-band padding and Laplace smoothing.
- (5) **Advance 2D Simulation (Step 6).** We advance the canonical wave state (η, ψ) under the Zakharov Hamiltonian via the integrating-factor AB2 scheme (15)–(16) (Section 3), producing the post-advance state $(\tilde{\eta}_{2D}^{n+1}, \tilde{\psi}_{2D}^{n+1})$.
- (6) **Forward DNO Map (Step 7).** We evaluate the post-advance surface vertical velocity $v = G(\tilde{\eta}_{2D}^{n+1}) \tilde{\psi}_{2D}^{n+1}$ via the forward DNO map (18) (Section 3.5).
- (7) **Apply Relaxation (Step 8).** We nudge $\tilde{\eta}_{2D}^{n+1}$ toward η^{target} via the exact exponential decay (34) on η alone, pointwise (Section 5.3), producing the final height field η_{2D}^{n+1} .
- (8) **Reconstitute ψ (Step 9).** We recompute ψ_{2D}^{n+1} from (η_{2D}^{n+1}, v) through the inverse Dirichlet–Neumann map so that the canonical pair $(\eta_{2D}^{n+1}, \psi_{2D}^{n+1})$ is consistent with the relaxed height (Section 5.3).
- (9) **Embed FAB Height Field (Step 10).** We set the level-set values of the 3D solver in the FAB ring from the post-relax 2D height field η_{2D}^{n+1} via $\phi = y - \eta$ (see Section 4.1); the interior of ϕ_{3D} is left untouched.
- (10) **Compute Volume Velocity from 2D (Step 11).** We reconstruct the bulk velocity at each depth layer from the canonical 2D state using the Craig–Sulem lift to $y = 0$ and the finite-depth harmonic kernels, as described in Sections 4.2–4.4 and summarized in Alg. 1.
- (11) **Interpolate Velocity in FAB from Depth Layers (Step 12).** We trilinearly interpolate the velocity within the FAB ring from the depth-layered velocities.
- (12) **Advance 3D (Step 13).** We follow standard NB-FLIP procedures to advance the 3D simulation, using the FAB-ring velocities as boundary conditions during pressure projection.

8 Validation

Experimental Setup. All experiments are conducted on a desktop with an NVIDIA GeForce RTX 4090. Implementation details of the 2D and 3D solvers are presented in Section 6. Relevant parameters and runtime performance are summarized in Table 5. In the wave-propagation benchmark and the Stokes-wave accuracy study we report both HOS-2 and HOS-3, while all other experiments use HOS-2, since this level is sufficient in the vast majority of our scenes and Order-2 is roughly $2\times$ faster than Order-3 per step (Table 3).

8.1 Comparisons and Validation Studies

Wave Propagation. To evaluate the accuracy of our wave model and our padding strategy, we simulated wave propagation in a square $20\text{ m} \times 20\text{ m}$ tank under shallow (1 m) and deep (20 m) water conditions with reflective boundary conditions on all four sides.

Algorithm 2 One substep of our coupled solver

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1:  $\Delta t \leftarrow \text{ComputeTimeStep}()$  ▷ outer 3D CFL
2:  $\tilde{\phi}_{3D}^{n+1}, \tilde{u}_{3D}^{n+1} \leftarrow \text{Advect3D}(\Delta t, u_{3D}^n)$ 
3:  $\mathcal{S}^{n+1} \leftarrow \text{UpdateSolid}(\Delta t, \mathcal{S}^n)$  ▷ rigid-body pose
4:  $\tilde{\eta}_{\text{samp}3D} \leftarrow \text{Sample3DHeight}(\tilde{\phi}_{3D}^{n+1})$  ▷ Sec. 5.1
5:  $\eta_{\text{target}} \leftarrow \text{Smooth}(\tilde{\eta}_{\text{samp}3D})$  ▷ Laplace smoothing
6:  $\tilde{\eta}_{2D}^{n+1}, \tilde{\psi}_{2D}^{n+1} \leftarrow \text{Advance2D}(\Delta t, \eta_{2D}^n, \psi_{2D}^n)$  ▷ Sec. 3
7:  $v \leftarrow \text{ForwardDNO}(\tilde{\eta}_{2D}^{n+1}, \tilde{\psi}_{2D}^{n+1})$  ▷ forward DNO, Eq. (18)
8:  $\eta_{2D}^{n+1} \leftarrow \text{ApplyRelaxation}(\Delta t, \tilde{\eta}_{2D}^{n+1}, \eta_{\text{target}})$  ▷ Sec. 5.3
9:  $\psi_{2D}^{n+1} \leftarrow \text{ReconstitutePsi}(\eta_{2D}^{n+1}, v)$  ▷ inverse DNO, Eq. (19)
10:  $\phi_{3D}^{n+1} \leftarrow \text{EmbedFABHeight}(\eta_{2D}^{n+1}, \tilde{\phi}_{3D}^{n+1})$  ▷  $\phi = y - \eta$  (FAB)
11:  $u_{\text{depth}}^{n+1} \leftarrow \text{ComputeVolumeVelFrom2D}(\eta_{2D}^{n+1}, \psi_{2D}^{n+1})$  ▷ Alg. 1
12:  $u_{\text{FAB}} \leftarrow \text{InterpDepthLayerFAB}(u_{\text{depth}}^{n+1})$ 
13:  $u_{3D}^{n+1} \leftarrow \text{Advance3D}(\Delta t, u_{\text{FAB}}, \tilde{u}_{3D}^{n+1}, \phi_{3D}^{n+1})$ 

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In both cases the surface is initialized at rest ($v = 0$) with a radially symmetric Gaussian height bump centered at the tank center, $\eta_0(x, z) = A \exp(-r^2/2\sigma^2)$ for $r < R$ and zero otherwise, where $r = \sqrt{(x - L/2)^2 + (z - L/2)^2}$, with width $\sigma = 1$ m and truncation radius $R = 6$ m; the peak amplitude is $A = 1$ m in the shallow setting and $A = 2$ m in the deep setting. Each configuration is propagated for 2000 frames at $\Delta t = 0.01$ s. As shown in Figure 13, we compared a total of four 2D wave models, i.e., the SWE [Chentanez and Müller 2010] and the BEM [Da et al. 2016] used in previous coupling methods [Chentanez et al. 2015; Huang et al. 2021; Thürey et al. 2006], the Airy Dispersion Kernel (DK) model [Canabal et al. 2016], and our nonlinear wave model, against a high-resolution 3D Eulerian level-set simulation that serves as the ground-truth reference. We found that our nonlinear wave model produces results most consistent with the 3D reference, and correctly captures wave reflection.

We quantify these differences in Table 4. For a fair comparison, every method’s output is resampled onto the same 128×128 height field before being compared with the reference; for BEM, which represents the surface as a mesh, we cast vertical rays downward from this 128×128 grid and take the first ray–mesh intersection as the per-cell height. For each frame we compute the absolute height error against the reference and reduce it to its spatial mean and max; we then average each of these quantities over the 2000 frames of the simulation, yielding the time-averaged mean and max errors reported in the table. Compared with SWE and BEM, our model reduces the time-averaged mean error by roughly 4× in the shallow-water case and by 3–5× in the deep-water case; even compared with the strongest baseline (Airy DK), it still cuts the mean error by about 2.4× in the shallow-water case and 1.7× in the deep-water case, with similar improvements on the max error. We also tested a strengthened Airy DK baseline that multiplies the dispersion relation by the amplitude-dependent factor $(1 + a^2 k^2)$. The correction improves the Airy DK baseline (row “Airy DK + dispersion fix” in Table 4), lowering its mean error by roughly 10% in both regimes, but the corrected baseline still incurs about 1.5–2.1× our mean error. Runtime comparisons among SWE, BEM, Airy DK, and our Order 2 and 3 models are listed in Table 3; our wave model runs about 1000× faster than BEM (though it could be further optimized). While

Table 3. Runtime comparison of the wave propagation experiment.

Method	Deep Water Time (ms)	Shallow Water Time (ms)
BEM	8693.7	25973.8
SWE	1.355	1.375
Airy DK	1.081	1.157
Ours (Order 2)	6.352	6.167
Ours (Order 3)	13.52	13.43

Table 4. Wave height error comparison of the wave propagation experiment. For each method, the first row is the mean error and the second row is the max error, each obtained by computing the spatial mean (resp. max) of the absolute height error on a 128×128 grid at every frame and then averaging over the 2000 frames of the simulation.

Method	Error	Shallow	Deep
SWE [Chentanez and Müller 2010]	Mean	0.048202	0.096199
	Max	0.210926	0.422074
BEM [Da et al. 2016]	Mean	0.051720	0.170493
	Max	0.234508	0.466547
Airy DK [Canabal et al. 2016]	Mean	0.027666	0.054496
	Max	0.126102	0.298801
Airy DK + dispersion fix	Mean	0.024655	0.049221
	Max	0.113956	0.280982
Ours (Order 2)	Mean	0.013607	0.032774
	Max	0.057469	0.174430
Ours (Order 3)	Mean	0.011501	0.034247
	Max	0.051843	0.161604

our model is more expensive than the linear-wave-based SWE and Airy DK solvers, it delivers the accuracy improvements reported in Table 4 and remains substantially cheaper than BEM. The 2D grid spacings used were $\Delta x = \frac{L}{128}, \frac{L}{128}, \frac{L}{128}$, and $\frac{L}{64}$ for the SWE, Airy DK, our wave model, and BEM, respectively, matching the grid spacing ratios used in Huang et al. [2021], while the 3D Eulerian reference uses a much finer grid spacing of $\Delta x = \frac{L}{256}$.

To further probe the role of the nonlinear correction, we sweep the nonlinearity parameter ε from 0 (linear Airy) to 1 (fully nonlinear HOS) in the shallow-water setting, using both the Order 2 and Order 3 HOS truncations, and measure the wave-height error against the same 3D reference. As shown in Figure 14, the error decreases monotonically as ε grows, confirming that the higher-order HOS terms are correctly oriented and pull the solver toward the 3D reference rather than away from it. The drop between $\varepsilon = 0$ and $\varepsilon = 0.1$ is already substantial, indicating that even a mild amount of nonlinearity removes a large fraction of the linear-model error. Order 2 and Order 3 are nearly indistinguishable for $\varepsilon \lesssim 0.6$, but the two curves separate beyond this point, with Order 3 attaining a noticeably lower error at $\varepsilon = 1$.

Stokes Wave Accuracy under Varying Nonlinearity. We assess the accuracy of our HOS wave solver against the analytical second-order Stokes wave (with the third-order frequency correction) on a

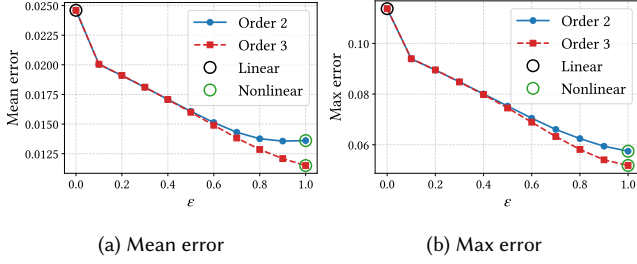


Fig. 14. Wave height error in the shallow water wave propagation experiment, plotted against the 3D Eulerian reference as a function of the nonlinearity parameter ε , for both Order 2 and Order 3 HOS truncations. Hollow black circles mark the linear case ($\varepsilon = 0$); hollow green circles mark the fully nonlinear case ($\varepsilon = 1$). The error decreases monotonically as ε grows, confirming that the higher-order HOS terms are correctly oriented; even a small ε already removes a large fraction of the linear-model error. Order 2 and Order 3 are nearly indistinguishable for $\varepsilon \lesssim 0.6$ and diverge beyond, with Order 3 attaining a noticeably lower error at $\varepsilon = 1$.

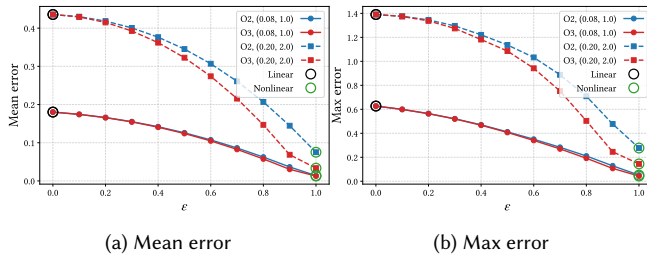


Fig. 15. Stokes wave propagation error against the analytical second-order Stokes solution as a function of the nonlinearity parameter ε . Four curves correspond to the two HOS truncation orders (O2, O3) and the two Stokes initial conditions parameterized by (ka, kh) . Hollow black circles mark the linear endpoint ($\varepsilon = 0$); hollow green circles mark the fully nonlinear endpoint ($\varepsilon = 1$).

periodic 1D domain of length $L = 2\pi$ through two complementary tests. Explicitly, the reference is the classical second-order Stokes profile [Dean and Dalrymple 1991]

$$\eta_{St}(x, t) = a \cos \theta + \frac{ka^2 (3 - \tanh^2(kh))}{4 \tanh^3(kh)} \cos 2\theta$$

with $\theta = kx - \omega t$, whose frequency follows the third-order Stokes dispersion relation [Whitham 2011] $\omega^2 = gk \tanh(kh) (1 + \delta)$ with

$$\delta = \frac{(ka)^2 (9 - 10 \tanh^2(kh) + 9 \tanh^4(kh))}{8 \tanh^4(kh)}.$$

The correction δ is the amplitude-dependent Stokes frequency shift, and dropping it recovers the linear relation $\omega^2 = gk \tanh(kh)$.

Long-time phase fidelity. We initialize four mild Stokes regimes, i.e., weakly nonlinear, non-breaking Stokes waves, with $(ka, kh) \in \{(0.08, 1.0), (0.10, 1.5), (0.12, 1.8), (0.13, 2.0)\}$ and propagate each for 18, 14, 12, and 10 linear periods, respectively. Figure 2 overlays the final-time surface elevation produced by the linear ($\varepsilon = 0$), intermediate ($\varepsilon = 0.5$), and fully nonlinear ($\varepsilon = 1$, HOS-3) solvers with the analytical Stokes profile. The linear solver (equivalent to

[Canabal et al. 2016]) progressively drifts out of phase across all four regimes because it lacks the amplitude-dependent Stokes frequency shift defined above, while the fully nonlinear HOS-3 solver remains phase-locked to the analytical curve. The intermediate $\varepsilon = 0.5$ curve lies between the two, but closer to the linear one.

Error sweep over ε . To examine how the nonlinearity parameter ε controls accuracy, we sweep $\varepsilon \in \{0, 0.1, \dots, 1.0\}$ across two HOS truncation orders (Order 2 and Order 3) and two distinct initial conditions with $(ka, kh) \in \{(0.08, 1.0), (0.20, 2.0)\}$. We propagate each configuration for ten linear periods and compare against the analytical Stokes solution frame by frame. The spatio-temporal error field $|\eta_{sim} - \eta_{Stokes}|/a$ is reduced to its mean and max, as shown in Figure 15. The error decreases monotonically as ε grows from 0 to 1, confirming that the higher-order DNO terms steer the linear solver toward the analytical Stokes wave. For the milder initial condition, Order 2 and Order 3 are nearly indistinguishable, indicating that the second-order kinematic boundary condition already captures the relevant nonlinear physics; for the steeper case, Order 3 noticeably outperforms Order 2, justifying the inclusion of the third-order term G_2 when handling steep waves.

Continuous Wavetrain Across the Coupling Interface. To stress-test the 2D–3D coupling under sustained periodic forcing, we drive a steady second-order Stokes wavetrain through an embedded Eulerian fluid box, as shown in Figure 3. A 1D wave domain of length 3 m and still-water depth 0.4 m hosts a 2 m×0.8 m fluid box centered at the origin. A wavemaker zone at the left edge of the wave domain softly relaxes (η, v) toward the analytical Stokes solution with steepness $ka = 0.3$ and wavelength 0.8 m, generating a right-going train; the entire right side of the wave domain acts as a sponge that absorbs outgoing waves before they wrap around the periodic boundary. The fluid box is initialized with still water and exchanges (η, ψ) with the 1D solver through the relaxation-zone protocol.

We use this setup to contrast the linear ($\varepsilon = 0$, equivalent to [Canabal et al. 2016]) and fully nonlinear ($\varepsilon = 1$) wave models when both are coupled to the same 2D Eulerian fluid box. Figure 3 shows three representative snapshots at $t \approx 2.65$ s, 3.75 s, 5.0 s (about 3.7, 5.2, and 7.0 wave periods). The red boxes mark the wave free-propagation region, where the 1D wave solver carries the train on its own. With the linear model this region develops visible kinks, the surface is no longer smooth, and the Stokes crest–trough asymmetry is entirely missing. The nonlinear model, by contrast, produces a smooth Stokes train whose shape inside the red boxes is continuous with the surface shape inside the 2D fluid box, with no visible discontinuity at the coupling interface. This indicates that the natively nonlinear 2D NS solver is incompatible with the 1D linear Airy wave model at this steepness ($ka = 0.3$): only the HOS wave model can match the 2D solver’s surface dynamics across the coupling; for gentler, nearly linear wavetrains the two models would agree and the cheaper Airy model would suffice. We note that this 1D setting has no transverse geometric spreading. In a full 2D wave domain, localized outgoing disturbances generally weaken as their wavefront expands, reducing nonlinear effects with distance.

Dispersion Matching Test. The 2D and 3D simulations may exhibit distinct numerical viscosity and dispersion characteristics. As noted by [Schreck and Wojtan 2022], such differences can potentially lead



Fig. 16. Dispersion matching test. The wave propagates from the 3D domain (bounded by dotted lines) into 2D regions. The wave retains a symmetric shape throughout its propagation, demonstrating consistent dispersion relations between the 3D and 2D simulations under our two-way coupled scheme. The wave model uses $\varepsilon = 1$.

to asymmetric wave propagation and spurious reflections at the coupling boundary. To investigate this, we conducted an experiment in a 5 m water tank with a depth of 1 m, where a wave packet with a peak amplitude of 0.2 m was initialized inside the 3D domain. The results are shown in Figure 16, demonstrating strong agreement between our 2D and 3D models: the wave maintained a highly symmetric shape throughout its propagation, and no artificial reflections were observed at the 2D–3D interface.

To highlight the importance of dispersion matching, we compared our method against three prior coupling schemes in Figure 6: Chentanez et al. [2015], Schreck and Wojtan [2022], and FAB [Stomakhin and Selle 2017] combined with the Airy DK wave model [Canabal et al. 2016]. In this combined baseline, the outer 2D wave field is advanced with the Airy DK model [Canabal et al. 2016], the 2D-to-3D direction of the coupling follows the FAB scheme [Stomakhin and Selle 2017], and the 3D-to-2D direction is handled with the procedure of Chentanez et al. [2015]; since the 3D-to-2D step is essentially standard and shared across most coupling methods, we label this baseline simply as FAB + Airy DK. All methods are driven by the same 3D NB-FLIP solver under identical initial conditions. The method of Chentanez et al. [2015] develops pronounced left-right asymmetry. For Schreck and Wojtan [2022], we recalibrate their dispersion-law rescaling factor for our 3D solver configuration and find that $d = 0.8$ gives the best result. This largely removes the left-right asymmetry, although the 2D and 3D regions remain visibly less consistent than with our method. We note that Schreck and Wojtan [2022] report no asymmetry under their original simulation settings. The FAB + Airy DK combination shows no visible asymmetry, but produces visibly deeper waves inside the 3D region than in the surrounding 2D field. Our method yields a wave field that is virtually identical inside and outside the 3D region, with a smoother 3D region. In terms of runtime, the wave-solver cost in this comparison is 13.3 ms per substep for our method, 27.5 ms for Schreck and Wojtan [2022], 3.7 ms for FAB + Airy DK, and 2.3 ms for Chentanez et al. [2015].

Crown Splash. A water ball with a radius of 0.05 m falls from a height of 0.2 m into a tank measuring 1.2 m wide and 0.6 m deep, as shown in Figure 8. We compare our coupling scheme against a 3D NB-FLIP reference and Chentanez et al. [2015]. We apply reflected padding to the boundaries of the 2D solver to enforce solid boundaries. Our coupling scheme produces results nearly identical to the NB-FLIP ground truth, enabling smooth wave transmission across the 2D–3D interface and accurate wave propagation throughout the

whole domain. The method of Chentanez et al. [2015], on the other hand, shows clearly incorrect wave propagation in the 2D region: since it relies on the shallow water equations, it fails to capture the dispersive wave dynamics.

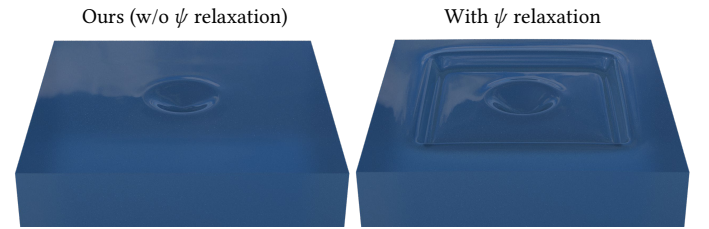


Fig. 17. Ablation on relaxing ψ , on the Crown Splash setup. The right panel additionally relaxes ψ toward a target obtained from a Helmholtz decomposition of the 3D surface velocity field, while the left panel is our method. Relaxing ψ produces an obvious abnormal depression in the 3D region.

ψ Relaxation Ablation. On the same Crown Splash setup, we ablate the asymmetric source placement of Section 5.2 by additionally relaxing ψ toward a Helmholtz-decomposed target extracted from the 3D surface velocity field. As shown in Figure 17, this symmetric variant produces spurious surface artifacts inside and around the embedding box.

Filter and Time-Integrator Ablation. We further ablate two components of the wave solver. First, disabling the low-pass state filter (17) causes the Crown Splash simulation to quickly become unstable. Second, replacing the IF-AB2 integrator (Section 3.4) with a first-order integrating-factor (exponential) Euler step increases the mean/max wave height error in the shallow-water wave-propagation experiment from 0.011501/0.051843 to 0.012752/0.053980 (cf. Table 4).

Single Boat. We simulate a 4.8 m-long boat moving at 4 m/s across a 72 m×3 m×36 m water domain; Figure 19 shows a rendering of this scene produced by our method with whitewater, where the feather-like Kelvin wake (“feather-like” follows the terminology of Rabaud and Moisy [2013]) is carried continuously across the 2D–3D interface. To assess our coupling scheme, we benchmark it against six alternatives under identical conditions, as shown in Figure 12: (a) Chentanez et al. [2015], (b) the Airy DK wave model [Canabal et al. 2016] coupled via the FAB scheme [Stomakhin and Selle 2017], (c) our method with velocity relaxation, (d) our wave model combined with the coupling scheme of Chentanez et al. [2015], (e) our full

Table 5. Specifications and statistics of the simulations. “Simulation Substeps” refers to 3D substeps.

Case	Res 2D	Res 3D	Δx_{2D} [m]	Δx_{3D} [m]	Frame Count	Frame Time Step [s]	Simulation Substeps	T_{substep} [ms]	CFL	ϵ
Dispersion Matching	256×256	256×64×512	0.020	0.0078	200	1/80	1133	87.8	0.25	1.0
Crown Splash	128×128	128×256×128	0.0094	0.0063	300	1/80	665	58.4	1.0	0.8
Single Boat (NB-FLIP)	/	1152×96×576	/	0.063	200	1/20	1273	363	1.0	/
Single Boat	576×288	384×96×192	0.13	0.063	200	1/20	1073	85.0	1.0	1.0
Seaplane	1024×256	768×128×256	0.156	0.047	240	1/20	3951	197.2	1.0	1.0
Battleship	512×256	512×256×256	0.234	0.047	240	1/20	3408	160.8	1.0	0.2
Submarine	512×256	512×128×256	0.156	0.047	360	1/30	2584	99.7	1.0	0.3
Two Chasing Boats	512×256	512×128×256	0.35	0.047	300	1/30	3396	268.8	1.0	1.0
Pond	1024×1024	512×256×512	0.0020	0.0018	300	1/30	14447	326.0	1.0	0.2

method, (f) a 3D NB-FLIP reference, and (g) a pure 3D Eulerian level-set reference. Method (a) produces a qualitatively wrong wake, as the SWE cannot resolve the dispersive Kelvin pattern. Methods (b), (c), and (d) reproduce the feather-like shape but exhibit clear spurious reflections behind the stern (highlighted by the red boxes in Figure 12). Our full method (e) reproduces the feather-like Kelvin wake with minimal spurious reflection at the 2D–3D interface. The NB-FLIP reference (f) closely matches our result, but its trailing wave at the far right is noticeably damped by the numerical dissipation of the 3D solver, while our coupled scheme preserves the wave because the 2D wave solver carries it without dissipation. The pure Eulerian level-set simulation (g) is included as an independent ground-truth reference. As reported in Table 5, our method also runs more than 4× faster than the pure GPU NB-FLIP simulation on this scene (85.0 ms vs. 363 ms per substep). We note that the feather-like appearance is a qualitative descriptor; the wake-angle study below provides the corresponding quantitative validation of the simulated wakes (Figure 18).

Specifically, according to [Rabaud and Moisy 2013], real-world ship wakes transition between the Kelvin angle and the Mach angle as the ship speed varies. We replicate this transition by simulating ship wakes at different speeds using the same experimental setting above. The resulting wake angles, illustrated in Figure 18, match the real-world data from [Rabaud and Moisy 2013] and correctly capture the Kelvin–Mach transition.

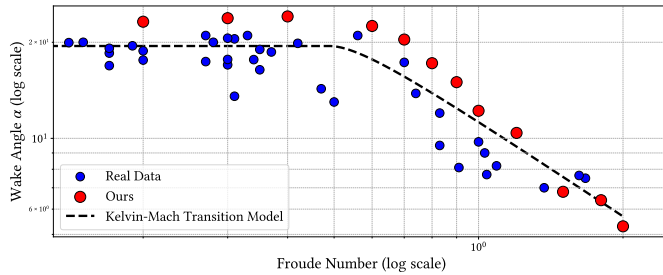


Fig. 18. Wake angle vs. Froude Number ($Fr = U/\sqrt{gL}$) compared with the Kelvin–Mach Transition Model [Rabaud and Moisy 2013]. All runs use $\epsilon = 1$.

8.2 Large-Scale 3D Demonstrations

Submarine. As shown in Figure 9, to demonstrate the capability of our coupling scheme in handling complex fluid–solid interactions, we simulate a submarine submerging below the sea surface and gradually rising up. The simulation domain spans 80 m×40 m for the sea surface, with an embedded 3D region of 24 m×6 m×12 m that includes a 10 m-long submarine moving at 4.3 m/s. As the submarine surfaces, it leaves a visible wake on the 2D ocean surface, while detailed 3D water motion is captured around the hull. To suppress nonlinear instabilities triggered by the rapidly moving hull, we cap the nonlinear amplitude parameter (see Section 3) at $\epsilon = 0.3$.

Seaplane. As shown in Figure 7, we simulate a seaplane landing on the ocean. The simulation domain spans 160 m×40 m for the sea surface, with an embedded 3D region of 36 m×6 m×12 m that includes a seaplane with a 10 m wingspan, pitched 9° nose-up and approaching at 8 m/s. Upon touchdown, the aircraft skims the surface at 10 m/s, leaving a long trailing wake that propagates smoothly across the 2D–3D interface.

Battleship in Heavy Waves. To further demonstrate the robustness of our coupling scheme, we simulate heavy ocean waves in a 120 m×60 m 2D domain and a 24 m×12 m×12 m 3D fluid domain. In this scenario, the battleship is simulated using weak two-way solid–fluid coupling. The results in Figure 11 show that our simulation remains stable even with heavy waves, and the 2D and 3D simulations remain visually consistent. Because the background spectrum drives the surface close to the breaking threshold, we further restrict the nonlinear amplitude parameter (see Section 3) to $\epsilon = 0.2$ to keep the wave evolution stable. This scene is our most demanding stress test, and the coupling is not entirely artifact-free: in the accompanying animation, wavefronts reflected from the rectangular boundary of the 3D box can be seen propagating outward into the 2D domain. Because the wave model is restricted to $\epsilon = 0.2$ for stability, it cannot fully absorb the strongly nonlinear surface content generated by the 3D solver under near-breaking forcing; the unabsorbed content instead radiates outward from the box boundary. Section 6 discusses practical mitigation strategies at render time, i.e., strengthening the ocean-spectrum displacement layer and emitting foam and whitewater around the box footprint.



Fig. 19. A 4.8 m-long boat cruising at 4 m/s across a 72 m×3 m×36 m water domain. The boat produces a clean feather-like Kelvin wake [Rabaud and Moisy 2013] whose long dispersive tail is carried far beyond the 3D fluid box by the 2D wave solver without visible discontinuity at the 2D–3D interface. The wave model uses $\varepsilon = 1$.

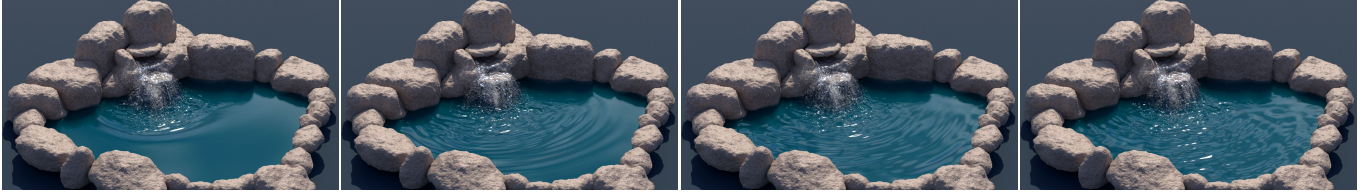


Fig. 20. Water falling into a shallow pond with irregular boundaries. The wave model uses $\varepsilon = 0.2$.

Two Chasing Boats. As shown in Figure 10, two boats with two independent 3D simulation domains are simulated on an ocean surface sized 180 m×90 m. The leading boat follows a curved path with one-way solid–fluid coupling. The trailing boat employs two-way solid–fluid coupling, responding dynamically to both the ambient waves and the wake generated by the leading boat.

Pond. We simulate water falling into a shallow pond with a depth of 0.3 m and a 2D domain spanning 2 m×2 m, with the embedded 3D fluid box covering 0.9 m×0.45 m×0.9 m. The pond outline and the rocks scattered inside it are encoded as an irregular solid mask on the 2D wave grid, while a separate rock mesh is treated as a rigid obstacle within the 3D fluid box. Figure 20 illustrates falling flow that generates splashes and ripples that propagate across the surface and interact with the irregular solid boundaries. Before every spectral wave step, the solver fills the masked cells by iteratively averaging the height η and vertical velocity v from their fluid-side neighbors, producing a discrete zero-gradient extension of (η, v) into the solid region. The subsequent FFT advance then sees this extension as replicated padding at the irregular boundary, so waves reflect off the pond walls and the rocks inside it at the cost of a few sweep iterations. Because the impact of the falling stream produces sharp local features that can excite high-wavenumber nonlinear instabilities, we cap the nonlinear amplitude parameter at $\varepsilon = 0.2$ to keep the wave evolution stable.

9 Conclusions and Limitations

Conclusions. We have presented a two-way coupled water solver that pairs a nonlinear and dispersive 2D wave model with a localized 3D Navier–Stokes solver, addressing a long-standing structural mismatch in graphics where the 2D side of every prior coupling pipeline was either linear (Airy) or non-dispersive (SWE). Our solver makes the opposite trade-off to prior 2D wave models: whereas they buy speed and robustness by discarding nonlinearity, and remain the better choice when the surface stays gentle and nearly linear, we accept a modestly more expensive and less stable wave step in

exchange for greater nonlinear accuracy and coupling fidelity that the strongly nonlinear scenes motivating a 2D–3D decomposition demand. The 2D model evolves the canonical Zakharov pair (η, ψ) under the exact water-wave Hamiltonian, advancing the Dirichlet–Neumann operator through a homogeneous Craig–Sulem expansion so that every nonlinear correction reduces to FFTs and pointwise products on a regular grid at $O(N^2 \log N)$ cost. Because the wave side and the 3D bulk solver carry the same physical variable, the surface velocity potential, the coupling becomes natural: both directions of state transfer are realized as Dirichlet–Neumann maps, and a Newtonian relaxation source integrated through its exact exponential decay returns the 3D-side surface state to the canonical wave variables without damping outgoing waves. The Hamiltonian structure of the wave model thus pays off chiefly in the coupling: the canonical pair (η, ψ) serves as a complete interface state that both solvers can read and write consistently. A scalar amplitude parameter $\varepsilon \in [0, 1]$ continuously interpolates between the linear Airy limit and the nonlinear Zakharov system, giving the user a single dial to trade nonlinear fidelity against stability. We validate our method on a wide range of experiments, including standalone 2D wave-propagation benchmarks, controlled 2D–3D coupling tests, and a suite of large-scale production-style scenarios such as a surfacing submarine, a seaplane landing, a battleship in heavy seas, two chasing boats, and a shallow pond with irregular boundaries. Across all these cases, our coupled pipeline produces accurate results with minimal coupling artifacts at competitive runtime, suggesting that the method is practical and applicable across a broad range of scenarios within its intended nonlinear regime.

Limitations. Our solver has several limitations. First, the fully nonlinear setting ($\varepsilon = 1$) is not unconditionally stable: when strong 3D forcing pushes the surface close to the breaking threshold, the higher-order DNO terms can excite numerical instabilities, and we have to reduce ε to maintain a stable simulation. In our experiments, the submarine uses $\varepsilon = 0.3$, while the battleship in heavy waves and the pond both use $\varepsilon = 0.2$. While our ε sweeps (Figure 14, Figure 15)

confirm that even a small ε already removes a large fraction of the linear-model error, the user is still left to tune ε per scene rather than always running at full nonlinearity. Second, the truncated DNO introduces a computational overhead that grows with the truncation order: as reported in Table 3, Order 3 is roughly $2\times$ slower than Order 2 per step, and each additional order adds a comparable cost. More broadly, even the Order-2 wave step is roughly $5\times$ more expensive than linear-wave solvers such as SWE and Airy DK (Table 3); in our coupled pipeline this overhead is minor in absolute terms, since the 3D solver dominates the total cost in most of our scenes (Table 5), but for standalone ocean animation where a linear model suffices, our solver is the more expensive choice. In practice, we find that Order 2 is already sufficient for the vast majority of our scenes, with Order 3 providing only marginal accuracy gains in the steepest regimes (Figure 15). Third, our 2D wave model cannot represent breaking waves: once the surface overturns, the single-valued height field $\eta(x, t)$ ceases to be a valid representation. We also note that the wave solver advances the entire 2D grid at every substep, so unlike Tessendorf-style spectral oceans, the simulated wave domain cannot be extended indefinitely. Finally, our coupling does not fully eliminate interface artifacts in the most demanding scenarios, such as the battleship scene, and we currently rely on render-time masking to visually mitigate these residual artifacts (Section 6).

Future Work. Each limitation points to a distinct direction. To address the conditional stability of the fully nonlinear model, we would like to investigate more stable nonlinear time-integration schemes for the Zakharov system, such as Hamiltonian-preserving integrators, spectrally consistent dealiasing tailored to the higher-order DNO terms, or implicit treatments of the leading nonlinear corrections, with the goal of pushing the stable operating point of ε closer to 1 even in strongly forced regimes. To reduce the per-step cost of the nonlinear correction, we will explore cheaper approximations of the higher-order DNO terms, for example mixed-precision FFT pipelines, learned surrogates for $G_j(\eta)$ at $j \geq 2$, or adaptive-order schemes that locally raise the truncation order only where the surface is steep. Finally, to extend the method to breaking and overturning waves, we plan to swap out the single-valued height-field representation in regions where breaking is detected for a richer surface representation, such as a locally multi-valued surface, a particle-augmented free surface in the spirit of NB-FLIP, or a hybrid Hamiltonian/Lagrangian description that retains the spectral efficiency of our solver in the smooth bulk while gracefully degrading to a particle representation through the breaking event.

Acknowledgments

We sincerely thank the reviewers for their valuable feedback. Georgia Tech authors acknowledge NSF CAREER #2420319, IIS #2433307, OISE #2433313, IIS #2433322, ECCS #2318814, and the NVIDIA Academic Grant for funding support. We credit the Houdini education license for video animations.

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